THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 10: **Circles** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Tangents PA and PB are drawn from a point P, 50 cm from the centre O of a circle of radius 30 cm. The length of each tangent is 40 cm. The length of the chord AB is: 1

(a) 80 cm **(b)** 48 cm **(c)** 60 cm **(d)** 24 cm

2. PA and PB are tangents from P to a circle, each of length 7 cm. A third tangent touches the circle at E on the minor arc AB and meets PA at C and PB at D. The perimeter of $\triangle PCD$ is: 1

(a) 14 cm **(b)** 9 cm **(c)** 7 cm **(d)** 21 cm

3. Tangents QA and QB are drawn from an external point Q to a circle, and $QA = (15x + 4)$ cm and $QB = (8x + 25)$ cm. The length of QA is: 1

(a) 51 cm **(b)** 49 cm **(c)** 3 cm **(d)** 70 cm

4. Two concentric circles have radii 68 cm and 32 cm. A chord of the larger circle touches the smaller circle. The length of the chord is: 1

(a) 60 cm **(b)** 120 cm **(c)** 100 cm **(d)** 72 cm

5. In $\triangle ABC$, right-angled at B, $AB = 45$ cm and $BC = 60$ cm. A circle is inscribed in the triangle. The radius of the inscribed circle is: 1

(a) 37 cm **(b)** 16 cm **(c)** 15 cm **(d)** 30 cm

6. A point P is 85 cm from the centre O of a circle of radius 77 cm. The length of the tangent drawn from P to the circle is: 1

(a) 162 cm **(b)** 8 cm **(c)** 36 cm **(d)** 1296 cm

7. AB is a chord of a circle with centre O and PA, PB are tangents. Then the line OP is: 1

(a) equal to AB **(b)** parallel to AB

(c) the perpendicular bisector of AB **(d)** a tangent to the circle

8. Tangents PA and PB are drawn from a point P, 50 cm from the centre O of a circle of radius 14 cm. The length of each tangent is 48 cm. The length of the chord AB is: 1

(a) 13.44 cm **(b)** 28 cm **(c)** 96 cm **(d)** 26.88 cm

9. **Assertion (A):** A tangent PT touches a circle with centre O at P. If $OP = 5$ cm and $PT = 12$ cm, then $OT = 13$ cm. 1

Reason (R): The tangent at any point of a circle is perpendicular to the radius through the point of contact.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** Exactly two tangents can be drawn to a circle from a point outside it. 1
Reason (R): A tangent to a circle intersects the circle at two points.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Tangents PA and PB are drawn from a point P, 75 cm from the centre O of a circle of radius 60 cm. The length of each tangent is 45 cm. Find the length of the chord AB. 2

12. PA and PB are tangents from P to a circle, each of length 9 cm. A third tangent touches the circle at E on the minor arc AB and meets PA at C and PB at D. Find the perimeter of $\triangle PCD$. 2

13. Two concentric circles have radii 90 cm and 72 cm. A chord of the larger circle touches the smaller circle. Find the length of the chord. 2

14. Tangents PA and PB are drawn from a point P, 75 cm from the centre O of a circle of radius 72 cm. The length of each tangent is 21 cm. Find the length of the chord AB. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. Tangents PA and PB are drawn from a point P, 50 cm from the centre O of a circle of radius 30 cm. The length of each tangent is 40 cm. Find the length of the chord AB. 3

16. In $\triangle ABC$, right-angled at B, $AB = 36$ cm and $BC = 77$ cm. A circle is inscribed in the triangle. Find the radius of the inscribed circle. 3

17. Tangents QA and QB are drawn from an external point Q to a circle, and $QA = (21x + 5)$ cm and $QB = (13x + 29)$ cm. Find the length of QA. 3

OR

Tangents PA and PB are drawn from a point P, 50 cm from the centre O of a circle of radius 14 cm. The length of each tangent is 48 cm. Find the length of the chord AB. 3

18. In $\triangle ABC$, right-angled at B, $AB = 40$ cm and $BC = 75$ cm. A circle is inscribed in the triangle. Find the radius of the inscribed circle. 3

OR

Tangents QA and QB are drawn from an external point Q to a circle, and $QA = (18x + 7)$ cm and $QB = (11x + 35)$ cm. Find the length of QA. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A circular park has centre O and radius 6 m. A gate P is at a distance of 10 m from O, outside the park. Two straight paths PA and PB touch the park at the points A and B. 5
(i) What is the measure of the angle between OA and the path PA?
(ii) Find the length of the path PA.
(iii) Write the length of the path PB.
(iv) Find the area of the quadrilateral OAPB.

20. Prove that in a right triangle ABC with $\angle B = 90^\circ$, the radius of the incircle is $r = \frac{AB + BC - AC}{2}$. 5

OR

Tangents PA and PB are drawn from a point P, 45 cm from the centre O of a circle of radius 36 cm. The length of each tangent is 27 cm. Find the length of the chord AB. 5

END OF QUESTION PAPER **Total: 40 Marks**